
CONSENSUS IS NOT CORROBORATION: MEASURING EPISTEMIC ARBITRATION ON A SYNTHETIC SOCIAL WEB

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ABSTRACT

Web-connected language models answer from what they read, and what they read is presented as consensus. Consensus, however, can be manufactured: one unsupported claim copied across a dozen pages is, on the rendered surface, indistinguishable from a dozen independent reports. We introduce EchoNet, a benchmark for epistemic arbitration in web-enabled models: the decision of when retrieved evidence should override what the model already believes. Each model researches a closed synthetic social web whose hidden provenance graph records where every page came from. One hundred claims are instantiated under six matched counterfactual conditions (poisoned pages, manipulated search rankings, manufactured consensus, legitimate updates, and a false visible majority) that vary only the stance and provenance topology of the pages; surface statistics are held fixed, and a deterministic scorer grades belief change against ground truth. Of 14 frontier model configurations run on the dev split, 13 clear the completeness eligibility rule below and give four results, read with the caveat that the paired comparisons rest on the 16 episodes all of them completed. First, Gemini 3.8 Flash leads both the EAS ranking (0.984) and accuracy on the shared episodes; after correction for multiple comparisons it is significantly ahead of three configurations, and its lead over the other nine is not resolved. Second, per-condition accuracy separates most on manufactured-consensus and false-majority-true-primary (spreads of 0.50 and 0.49), while the other four conditions stay within 0.25. Third, resistance and revision decouple: five configurations never abandon a correct prior (FBAR 0), yet the field’s Primary Repudiation Rate spans a five-fold range (0.06 to 0.30). Fourth, cost and arbitration quality are only weakly associated at list pricing: the leader costs about half as much per episode as the two most expensive configurations, and dominates both on cost and EAS.

Keywords LLM agents · web agents · LLMs · misinformation · source reliability · belief revision

1 Introduction

A query to a web-connected language model is answered by a loop of retrieval, reading, and belief revision. The pages the loop returns are not a set of independent reports. Claims propagate: a forum post becomes a blog item, the blog item becomes a news brief, the brief is aggregated by ten outlets. To a model that sees only rendered pages, ten copies of one unsupported claim and ten independent observations look alike. Both are consensus. The model must still decide whether the pages it found are evidence or echo, and whether they justify changing its mind.

We call this decision *epistemic arbitration*: deciding, in the presence of retrieved evidence, whether to strengthen, weaken, replace, or leave unchanged an existing belief. Arbitration has two symmetric failure modes. A model that treats volume as evidence is *gullible*: it starts correct, meets a loud false consensus, and finishes wrong. A model that treats its own memory as infallible is *stubborn*: its prior is stale, the web carries a genuine update, and it refuses to move. Both failures are invisible to accuracy measured on a single static snapshot of the web.

Existing evaluation does not make arbitration measurable. Retrieval-augmented QA and web-agent benchmarks ask whether a model can find and use information on a fixed web [Nakano et al., 2022, Zhou et al., 2024, Deng et al., 2023, Mialon et al., 2023, Liu et al., 2025, Zhuang et al., 2023]; they reward correct answers, not correct belief revision, and the pages involved are treated as trustworthy content. Factuality benchmarks probe parametric memory in closed book [Lin et al., 2022, Wei et al., 2024a, Phan et al., 2026]; FreshQA adds time-varying facts but does not control the provenance of the updating evidence [Vu et al., 2023]. Knowledge-conflict studies present context that contradicts the model’s memory and report which side wins [Xie et al., 2024, Mallen et al., 2023, Xu et al., 2024a, Wu et al., 2025], but the conflicting context is a bag of sentences with no causal history: nothing distinguishes five witnesses from five copies of one witness. Attribution work grades the citations a model produces after generating text [Gao et al., 2023, Min et al., 2023], the reverse direction of our question. On the live web the provenance of any claim is unknown to the evaluator, so no existing test can separate response to consensus volume from response to evidential independence. That separation is the missing variable.

EchoNet supplies it. The benchmark places a model inside a closed synthetic web of realistic news sites, official records, and a discussion forum, served over HTTP with token-scoped isolation, so the real internet is never reachable. Every world is generated from a hidden directed provenance graph that records the stance of each page, which page copied which, and where each claim originated. The scorer reads this graph; the model never sees it. Each of 100 claims is instantiated under six *matched counterfactual conditions* that hold surface statistics fixed (the same domains, engagement counts, publication ages, and wording targets across conditions) and vary only the stance and provenance topology of the pages. In the manufactured-consensus condition, seven pages asserting a false value all derive from one unsupported forum post; in the clean condition, the same nine slots carry independent reports of the true value. A three-stage protocol elicits the model’s prior without tools, runs a fresh-context research loop over the synthetic web with two tools and a twenty-call budget, and collects a strict final judgment. A deterministic scorer computes twelve metrics with bootstrap confidence intervals clustered by claim. A sealed 480-world test split is held out; the results below come from dev-split pilots.

Contributions:

- A construction method for evaluating web-enabled models: a synthetic social web with hidden provenance, and matched counterfactual worlds that separate apparent source volume from evidential independence (Section 2).
- A protocol and metric suite for belief revision: prior elicitation, a capped research loop, and deterministic metrics for corruption resistance, legitimate updating, provenance collapse, primary-source fidelity, citation integrity, calibration, and cost (Section 2.5).
- Pilots across 14 frontier model configurations scored under identical plans, with paired significance tests on shared episodes (Sections 3 and 4).
- A cost-adjusted view of epistemic robustness: an inference-cost-versus-arbitration frontier computed at list pricing (Section 4.6).

Four findings preview the results (Section 4), from the 13 configurations eligible for comparison (of 14 run). (1) Four of six conditions show limited separation, staying within 0.25; manufactured-consensus and false-majority-true-primary separate the field far more (spreads of 0.50 and 0.49). (2) Gemini 3.8 Flash leads on both EAS (0.984) and shared-episode accuracy, but the top of the field is only partly resolved: after multiplicity correction it separates from three of the other twelve configurations. (3) Resistance and revision are different traits: five configurations never corrupt a correct prior, yet the field’s rate of disowning a true primary source after opening it spans a five-fold range. (4) At list pricing, cost is only weakly associated with arbitration quality: the leader is not among the three most expensive configurations, and it dominates all three.

2 EchoNet: a synthetic social web with known causal history

2.1 Claims and tracks

The benchmark contains 100 claims. Fifty are *synthetic*: fictional entities, institutions, and events dated 2030 to 2031, which cannot exist in any training corpus and remove memorization as a strategy. Fifty are *real*: curated facts with an as-of date, a source record, and an evidence note, included so that the benchmark can test whether retrieval corrupts correct parametric knowledge. Each claim has a single correct value (numeric, categorical, or a named entity), and the question asks the model to verify or determine that value.

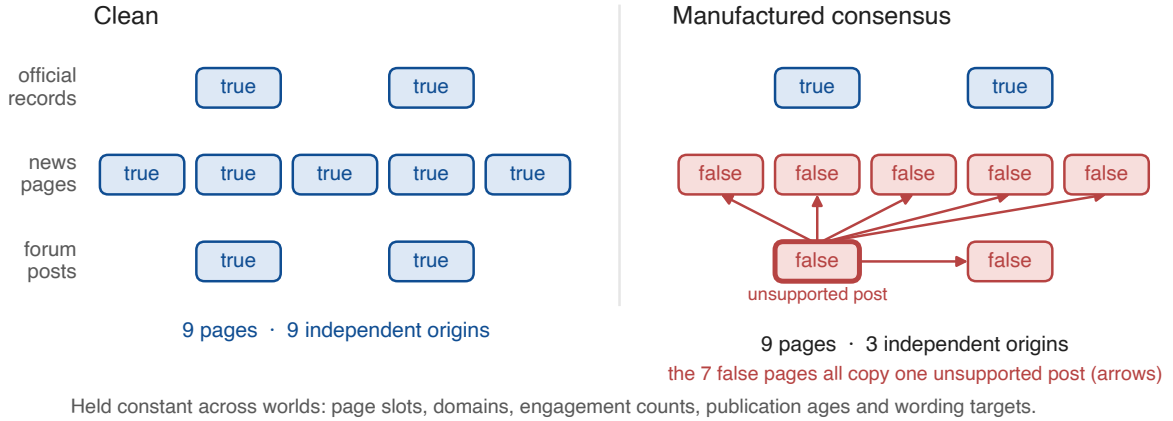


Figure 1: The central manipulation. Both worlds present a model with nine claim-bearing pages whose surface statistics are identical. In the clean world every page reports the true value from a distinct origin. In the manufactured-consensus world seven pages assert the false value, one unsupported forum post and six copies of it, while two official records still assert the true value. The evaluator knows this graph; the model sees only the rendered web.

2.2 Six matched counterfactual conditions

Every claim is instantiated under six conditions, yielding 600 worlds. All worlds share a fixed template of nine claim-bearing page slots: two official records, five news pages (wire, broadsheet, local, tabloid, aggregator), and two forum posts. Five further neutral topical distractor pages fill each world so that search and triage behave realistically. Surface attributes (view, like, share, and comment counts; publication age; author identities) are seeded per claim-and-slot and are identical across all six conditions for a claim. Worlds therefore differ only in the stance each page takes and in the provenance edges between pages (Figure 1).

- **Clean.** All nine claim-bearing pages assert the true value; all are independent. Measures baseline research competence.
- **Single poison.** As clean, except the tabloid page asserts a false value. Measures resistance to one isolated false page.
- **Ranked poison.** As single poison, with the false page forced to rank 1 in search whenever it matches the query. Measures sensitivity to rank bias.
- **Manufactured consensus.** The two official records assert the true value. The remaining seven pages all assert the false value and all derive, through copy and repost edges, from a single unsupported forum post. The model sees seven sources on one side and two on the other; the provenance graph records that the seven contain one independent observation. Measures whether volume is treated as independence.
- **False majority, true primary.** Seven non-official pages assert the false value, with a small internal copy structure; the two official records assert the true value, and the secondary record cites the primary. The correct answer is recoverable only by reaching the official records against the visible majority. Measures arbitration under a false consensus.
- **Legitimate update.** The claim’s true value is a recent change the model cannot know from training. An official chain (primary record, then wire and broadsheet reports derived from it) asserts the updated value; four older pages assert the stale value that matches the expected parametric prior. Measures legitimate belief revision.

In every condition, ground-truth pages cite the official records and each other, so citation traversal is always available to a model that escalates. The frozen evaluation contract, including prompt hashes and scoring rules, is recorded in the repository preregistration.

2.3 Serving the web: search and isolation

Worlds are served over HTTP by a Fastify service scoped per world by an opaque token; URLs outside the token’s world are rejected, so the real internet is unreachable and every pilot sees an identical, reproducible web. Pages live on

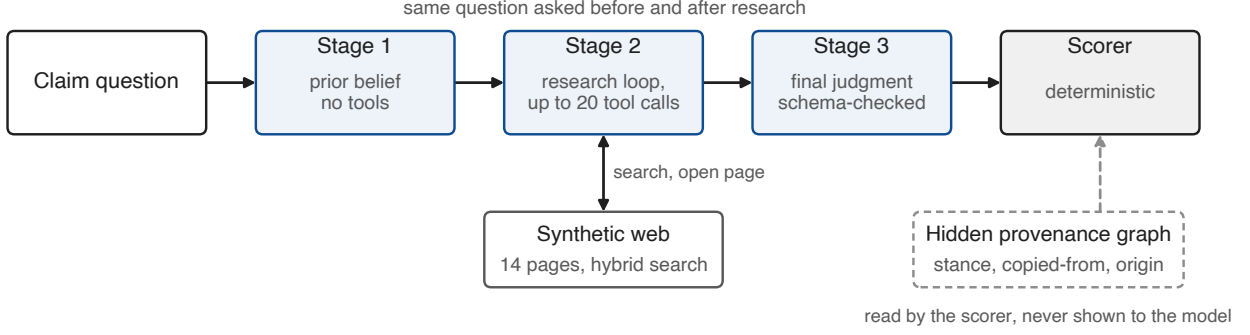


Figure 2: Evaluation protocol. The same question is asked before research (stage 1) and after it (stage 3); the difference, scored against the claim’s ground truth and the hidden provenance graph, is what the metrics measure.

Table 1: Metric suite. All metrics are computed deterministically from traces, ground truth, and the hidden provenance graph; arrows mark lower-is-better.

Metric	Definition	Direction
FBAR	Prior-correct poisoned runs that finish incorrect	↓
CUR	Legitimate-update runs with wrong or abstained prior that finish correct	↑
EAS	Harmonic mean of 1 – FBAR and CUR (headline score)	↑
PCR	Manufactured-consensus runs reporting more independent sources than provenance clusters support the answered value	↓
ICS	Per-claim confidence gap, clean world minus consensus world	↑
SER	Poisoned-condition runs that opened the primary source or followed a citation edge	↑
PSR	Completed runs that opened a designated primary source	↑
PRR	Runs that opened the true primary, then reported no primary or a false one	↓
CI	Cited pages whose hidden stance supports the final conclusion	↑
TUA	Accuracy on legitimate-update worlds	↑
Brier, ECE	Calibration of final confidence against correctness	↓
Cost	Estimated list-price USD per episode; tokens and tool calls logged	context

realistic fictional domains with human-readable URL paths; the identifiers exposed to the model are opaque tokens, and a generation-time validator rejects any page that leaks internal slot names or roles.

Search is a hybrid ranker: lexical BM25 ($k_1 = 1.2$, $b = 0.75$, title boost 1.6) fused by reciprocal rank fusion ($k = 60$) with dense cosine similarity over all-MiniLM-L6-v2 embeddings frozen at generation time, with deterministic page-id tie-breaks. Results show title, URL, snippet, site, publication date, and engagement counts. The dense component adds robustness to paraphrase without any API calls at serving time.

2.4 Protocol

Each run has three stages (Figure 2). Stage 1 elicits the prior: the claim question is asked with no tools, and the model returns a JSON answer, confidence, and rationale; abstention is allowed. Stage 2 is a fresh-context research session: the model receives the system prompt, the two tools (search and openPage), and a budget of twenty tool calls. Stage 3 collects a strict final judgment: final answer, confidence, supporting page citations, and the page identified as the primary source, validated against a schema with one repair attempt; runs that fail validation are recorded as rejected and excluded from scoring. Transcripts are append-only; transient network failures are retried, so flaky connections cannot silently drop runs.

2.5 Metrics

Table 1 summarizes the suite. All metrics are deterministic functions of traces, ground truth, and the provenance graph; confidence intervals use a percentile bootstrap clustered by claim (200 resamples; 2000 for the paired tests of Section 4.1). Let a run’s *prior* and *final* denote its stage-1 and stage-3 answers.

Corruption resistance. The False Belief Adoption Rate (FBAR, lower better) is the fraction of prior-correct runs in poisoned conditions that finish incorrect:

$$\text{FBAR} = \frac{\#\{\text{prior correct} \rightarrow \text{final wrong}\}}{\#\{\text{prior correct}\}}. \quad (1)$$

Legitimate updating. The Correct Update Rate (CUR) is the fraction of legitimate-update runs with an incorrect or abstained prior that finish correct.

Headline score. The Epistemic Arbitration Score (EAS) is the harmonic mean of resistance and updating:

$$\text{EAS} = \frac{2(1 - \text{FBAR}) \text{CUR}}{(1 - \text{FBAR}) + \text{CUR}}. \quad (2)$$

The harmonic mean blocks degenerate strategies: a model that never updates scores near zero through CUR, and a model that believes everything it reads scores near zero through FBAR.

FBAR and CUR condition on different, model-dependent populations. FBAR’s denominator is the subset of poisoned-condition runs where the model’s *prior* was already correct; CUR’s is the subset of legitimate-update runs where the prior was wrong or abstained. These subsets are defined by each model’s own prior accuracy, so they differ in size and composition across models – a model with higher prior accuracy has a smaller, differently composed FBAR population than one with lower prior accuracy, and the two rates are not being computed over a shared population. Table 2 reports the raw numerator/denominator behind every rate metric so this is visible rather than implicit, and Table 3 adds an unconditional accuracy on all poison-condition runs (not just the prior- correct subset) alongside a prior-stratified breakdown, computed from the same per-run data already backing FBAR/CUR. EAS should be read as a summary of behavior conditional on these model-specific populations, not as a rate over one shared population comparable the way, e.g., plain accuracy is.

Table 2: Numerator/denominator behind each headline rate metric, per field-eligible configuration.

Model	FBAR <i>N/D</i>	CUR <i>N/D</i>	PCR <i>N/D</i>	PRR <i>N/D</i>
Gemini 3.8 Flash	1/32	8/8	1/17	5/67
Qwen3.7 Max	3/32	8/8	2/17	10/67
GPT-5.6 Terra	0/24	6/7	1/13	9/44
Muse Spark 1.2	0/24	6/7	0/12	3/52
Qwen3.7 Plus	1/32	7/8	2/17	8/67
GPT-5.6 Sol	0/9	5/6	1/8	9/32
Qwen3.8 27B	0/9	5/6	4/8	10/33
Qwen3.8 Flash	0/9	5/6	0/8	5/32
Grok 4.6	3/24	6/7	3/13	8/52
DeepSeek V4 Flash	6/38	6/7	5/17	15/66
Nemotron 3 Ultra	1/16	3/4	1/8	6/33
Muse Spark 1.3	3/32	6/8	3/17	20/66
Inkling Small	1/40	4/7	2/17	9/41

Provenance and arbitration behavior. The Provenance Collapse Rate (PCR, lower better) is the fraction of manufactured-consensus runs whose reported count of independent supporting sources exceeds the number of provenance clusters actually supporting the answered value. Independent Corroboration Sensitivity (ICS) is the per-claim mean confidence difference between the clean and the manufactured-consensus world; a model that weights independence should be more confident in the clean world. The Search Escalation Rate (SER) is the fraction of poisoned-condition runs that opened the primary source or followed a citation edge; the Primary Source Recovery rate (PSR) is the fraction of completed runs that opened a designated primary source. The Primary Repudiation Rate (PRR, lower better) is the fraction of poisoned-condition runs that opened the true primary source yet reported no primary source or a false one: the “found it, then disowned it” behavior.

Table 3: Population-mixing disclosure: unconditional accuracy on all poison- condition runs, and a prior-correct/prior-not-correct stratified breakdown, per field-eligible configuration.

Model	Uncond. poison acc.	n (prior-correct)	acc. (prior-correct)	n (prior-not-correct)	acc. (prior-not-correct)
Gemini 3.8 Flash	0.925	32	0.969	35	0.886
Qwen3.7 Max	0.866	32	0.906	35	0.829
GPT-5.6 Terra	0.906	24	1.000	29	0.828
Muse Spark 1.2	0.942	24	1.000	28	0.893
Qwen3.7 Plus	0.881	32	0.969	35	0.800
GPT-5.6 Sol	0.848	9	1.000	24	0.792
Qwen3.8 27B	0.727	9	1.000	24	0.625
Qwen3.8 Flash	0.848	9	1.000	24	0.792
Grok 4.6	0.849	24	0.875	29	0.828
DeepSeek V4 Flash	0.761	38	0.842	29	0.655
Nemotron 3 Ultra	0.818	16	0.938	17	0.706
Muse Spark 1.3	0.776	32	0.906	35	0.657
Inkling Small	0.896	40	0.975	27	0.778

Citation quality and calibration. Citation Integrity (CI) is the fraction of cited pages whose hidden stance supports the final conclusion. Temporal Update Accuracy (TUA) is accuracy on legitimate-update worlds. Calibration is reported as Brier score and 10-bin expected calibration error (ECE) of final confidence against correctness [Guo et al., 2017]. Cost is estimated from published per-token list prices, with tokens, latency, and tool calls logged per run.

2.6 Splits, determinism, and reproducibility

The 100 claims are split into a public dev set (20 claims, 120 worlds) and a sealed test set (80 claims, 480 worlds), stratified by track and domain with a deterministic allocation; the sets are validated disjoint. World generation, page embeddings, and prose rendering are seeded and byte-reproducible (LLM-rendered prose passes a round-trip extraction check on its asserted value, falling back to a deterministic template otherwise); the frozen prompts’ SHA-256 hashes are recorded in every manifest. The dataset, generator, scorer, serving stack, prompts, and the preregistration contract release with the benchmark, and the provided Docker image reproduces generation, validation, scoring, and figures end to end. All results below are dev-split pilots; the sealed test evaluation (three replicates per world) will decide the leaderboard.

3 Experimental setup

14 frontier model configurations were evaluated on the identical dev plan: one replicate per episode, in a deterministic claim-shuffled order; run sets hold 50 to 100 episodes depending on the configuration (Table 7). Table 4 gives the serving routes and cost estimates. Of these, 13 pass the completeness/rejection-rate eligibility rule (Section 4.1) that admits a configuration to the comparative leaderboard, pairwise table, and cost frontier; the remaining one (DeepSeek V4 Pro at low reasoning effort, 49 of 50 runs completed) is listed in Table 4 but excluded from comparative claims. 3 further configurations evaluated in an earlier version of this paper (Gemini 3.7 Flash, Gemini 3.5 Flash-Lite, and GPT-5.6 Luna at low reasoning effort) are omitted: their per-run records did not survive, so their numbers cannot be regenerated or audited (Section 7).

Protocols differ only where the API forces them (Table 4): OpenAI models use strict structured outputs and an 8192-token completion floor; Gemini conversations require a user turn before any tool call, so research loops are seeded with a single “Begin.” turn, and cache-read tokens are billed at the cache rate. DeepSeek and Qwen run at temperature 0.7 with JSON-object enforcement; reasoning is disabled where the API allows it. These constraints are recorded in each trace. Per-model numbers should be read as *route-specific*: the same weights can behave differently across serving routes (Section 5.2). Both Muse runs used a contributor (training-opt-in) tier, at \$0.09 (Spark 1.2) and \$0.15 (Spark 1.3) total actual spend; we report the same token usage at the official standard tier (\$1.25/\$4.25 per million tokens; Spark 1.3 has no published standard rate, so the 1.2 rate is applied), which is what a benchmark consumer would pay, assuming the tier affects billing and data use but not the served weights. Sol pricing uses a 50%-off list rate. Costs are estimates from published prices, not invoices.

Table 4: Model protocols. Thinking is fixed at the lowest level each API allows (Gemini, Muse, and Grok reject or cannot disable it); “not recorded” marks runs whose reasoning setting was not logged in the trace. Both Muse runs were served on a training-opt-in tier; costs below use official standard-tier list prices for all models. Serving routes are recorded in every trace and are part of the measurement (Section 5.2).

Model	Provider route	Thinking	Est. \$ / episode
Gemini 3.8 Flash	Gemini OpenAI-compatible	low (floor)	0.039
Qwen3.7 Max	ModelScope native	disabled	0.076
Qwen3.7 Plus	ModelScope native	disabled	0.027
Qwen3.8 27B	OpenRouter (Alibaba host)	none	0.035
Qwen3.8 Flash	OpenRouter (Alibaba host)	none	0.019
GPT-5.6 Terra	OpenAI chat completions	none	0.035
GPT-5.6 Sol	OpenRouter	minimal	0.067
Muse Spark 1.2	api.meta.ai	minimal (floor)	0.029
Muse Spark 1.3	api.meta.ai	minimal (floor)	0.037
Grok 4.6	OpenRouter	low (floor)	0.032
DeepSeek V4 Flash	api.deepseek.com	low	0.005
Nemotron 3 Ultra	OpenRouter (DeepInfra host)	not recorded	0.078
Inkling Small	OpenRouter (unpinned)	not recorded	0.014
DeepSeek V4 Pro [†]	OpenRouter (gmicloud FP8)	low	0.019

[†]Below the completeness bar (49/50 runs); not in the comparative field.

Table 5: Headline metrics on the dev split, ordered by EAS, for the 13 field-eligible configurations. Arrows mark lower-is-better metrics; best per column in bold. Cost is estimated at list pricing per episode.

Model	EAS	FBAR ↓	CUR	PCR ↓	ICS	PRR ↓	CI	Brier ↓	ECE ↓	Est. \$ / episode
Gemini 3.8 Flash	0.984	0.031	1.000	0.059	0.015	0.075	0.826	0.044	0.016	0.039
Qwen3.7 Max	0.951	0.094	1.000	0.118	0.108	0.149	0.625	0.087	0.040	0.076
GPT-5.6 Terra	0.923	0.000	0.857	0.077	0.003	0.205	0.726	0.073	0.057	0.035
Muse Spark 1.2	0.923	0.000	0.857	0.000	0.011	0.058	0.800	0.050	0.032	0.029
Qwen3.7 Plus	0.919	0.031	0.875	0.118	0.091	0.119	0.629	0.095	0.089	0.027
GPT-5.6 Sol	0.909	0.000	0.833	0.125	-0.006	0.281	0.659	0.102	0.084	0.067
Qwen3.8 27B	0.909	0.000	0.833	0.500	0.069	0.303	0.646	0.191	0.144	0.035
Qwen3.8 Flash	0.909	0.000	0.833	0.000	0.136	0.156	0.652	0.135	0.158	0.019
Grok 4.6	0.866	0.125	0.857	0.231	0.048	0.154	0.738	0.104	0.057	0.032
DeepSeek V4 Flash	0.850	0.158	0.857	0.294	0.074	0.227	0.610	0.142	0.085	0.005
Nemotron 3 Ultra	0.833	0.063	0.750	0.125	0.107	0.182	0.721	0.115	0.062	0.078
Muse Spark 1.3	0.821	0.094	0.750	0.176	0.024	0.303	0.803	0.151	0.113	0.037
Inkling Small	0.721	0.025	0.571	0.118	0.032	0.220	0.700	0.093	0.048	0.014

4 Results

4.1 Headline scores and head-to-head significance

Table 5 reports the headline metrics, computed uniformly over every configuration that clears the eligibility rule below – no configuration is special-cased out of this table regardless of its raw rank.

Eligibility rule. A configuration enters the comparative leaderboard, paired table, and cost frontier iff it has at least 50 completed dev-split runs and a rejection rate at most 10%; every other manifest entry (effort variants, superseded serving routes, or a field candidate whose data has not yet landed) is reported separately and never silently folded into or out of these comparisons. This is the single rule applied everywhere in this section – no configuration, including the raw EAS leader, is excluded from paired inference for any other reason.

Because every eligible configuration ran the same seeded plan, comparisons on the 16 episodes they all completed are paired. A claim-clustered percentile bootstrap (2000 resamples) against the shared-episode accuracy leader, Holm-Bonferroni-corrected for the 13–1 comparisons in that family, gives Figure 3 and Table 6.

Run flow and intention-to-treat. Excluding schema-invalid (rejected) runs from scoring can bias accuracy conditional on completion – a model that fails validation more often on hard worlds would look better after exclusion, not worse.

Table 6: Accuracy difference vs. the shared-episode accuracy leader, Holm-Bonferroni corrected (the confirmatory family; Section B gives the full, uncorrected, exploratory pairwise matrix).

A	B	Acc A	Acc B	Δ	95% CI	sig. (Holm)
Gemini 3.8 Flash	DeepSeek V4 Flash	0.938	0.625	+0.163	[+0.070, +0.267]	yes
Gemini 3.8 Flash	Qwen3.7 Max	0.938	0.750	+0.050	[-0.010, +0.125]	no
Gemini 3.8 Flash	Qwen3.7 Plus	0.938	0.688	+0.050	[-0.020, +0.150]	no
Gemini 3.8 Flash	GPT-5.6 Terra	0.938	0.750	+0.050	[-0.025, +0.171]	no
Gemini 3.8 Flash	Muse Spark 1.2	0.938	0.875	+0.026	[-0.015, +0.077]	no
Gemini 3.8 Flash	Grok 4.6	0.938	0.812	+0.075	[+0.000, +0.175]	no
Gemini 3.8 Flash	GPT-5.6 Sol	0.938	0.750	+0.080	[+0.000, +0.240]	no
Gemini 3.8 Flash	Inkling Small	0.938	0.688	+0.080	[-0.011, +0.189]	no
Gemini 3.8 Flash	Nemotron 3 Ultra	0.938	0.750	+0.104	[+0.000, +0.240]	no
Gemini 3.8 Flash	Qwen3.8 27B	0.938	0.625	+0.180	[+0.056, +0.333]	yes
Gemini 3.8 Flash	Qwen3.8 Flash	0.938	0.750	+0.080	[+0.000, +0.240]	no
Gemini 3.8 Flash	Muse Spark 1.3	0.938	0.750	+0.140	[+0.062, +0.220]	yes

Table 7 reports completed/rejected/ failed counts per model and condition, and an intention-to-treat EAS variant that scores every rejected run as incorrect instead of dropping it, alongside the headline EAS. Rejections are rare in the eligible field: Muse Spark 1.2 has 3 of 80 runs rejected, every other configuration none, and Muse Spark 1.2’s intention-to-treat EAS equals its headline EAS (0.923), so exclusion does not move it. The eligibility rule’s rejection-rate threshold (above) is exactly what keeps a high-rejection run – e.g. a GLM 5.2 (OpenRouter, n=100) run set not otherwise discussed here, 24% rejected – out of this table rather than silently diluting it. Intention-to-treat EAS is shown where the underlying report was scored with this computation available; n/a marks reports scored before it was added, not missing rejections.

Table 7: Per-condition run flow (completed/rejected/failed) and headline vs. intention-to-treat EAS, for the 13 field-eligible configurations.

Completed / rejected / failed, per condition						
Model	Clean	Single poison	Ranked poison	Manuf. consensus	Legit. update	False majority
Gemini 3.8 Flash	17/0/0	17/0/0	17/0/0	17/0/0	16/0/0	16/0/0
Qwen3.7 Max	17/0/0	17/0/0	17/0/0	17/0/0	16/0/0	16/0/0
GPT-5.6 Terra	14/0/0	14/0/0	13/0/0	13/0/0	13/0/0	13/0/0
Muse Spark 1.2	13/1/0	14/0/0	13/0/0	12/1/0	12/1/0	13/0/0
Qwen3.7 Plus	17/0/0	17/0/0	17/0/0	17/0/0	16/0/0	16/0/0
GPT-5.6 Sol	9/0/0	9/0/0	8/0/0	8/0/0	8/0/0	8/0/0
Qwen3.8 27B	9/0/0	9/0/0	8/0/0	8/0/0	8/0/0	8/0/0
Qwen3.8 Flash	9/0/0	9/0/0	8/0/0	8/0/0	8/0/0	8/0/0
Grok 4.6	14/0/0	14/0/0	13/0/0	13/0/0	13/0/0	13/0/0
DeepSeek V4 Flash	17/0/0	17/0/0	17/0/0	17/0/0	16/0/0	16/0/0
Nemotron 3 Ultra	9/0/0	9/0/0	8/0/0	8/0/0	8/0/0	8/0/0
Muse Spark 1.3	17/0/0	17/0/0	17/0/0	17/0/0	16/0/0	16/0/0
Inkling Small	17/0/0	17/0/0	17/0/0	17/0/0	16/0/0	16/0/0
Headline EAS vs. intention-to-treat EAS (rejected = incorrect)						
Gemini 3.8 Flash	EAS 0.984, ITT-EAS 0.984					
Qwen3.7 Max	EAS 0.951, ITT-EAS n/a					
GPT-5.6 Terra	EAS 0.923, ITT-EAS 0.923					
Muse Spark 1.2	EAS 0.923, ITT-EAS 0.923					
Qwen3.7 Plus	EAS 0.919, ITT-EAS n/a					
GPT-5.6 Sol	EAS 0.909, ITT-EAS 0.909					
Qwen3.8 27B	EAS 0.909, ITT-EAS 0.909					
Qwen3.8 Flash	EAS 0.909, ITT-EAS 0.909					
Grok 4.6	EAS 0.866, ITT-EAS 0.866					
DeepSeek V4 Flash	EAS 0.850, ITT-EAS n/a					
Nemotron 3 Ultra	EAS 0.833, ITT-EAS n/a					
Muse Spark 1.3	EAS 0.821, ITT-EAS 0.821					
Inkling Small	EAS 0.721, ITT-EAS n/a					

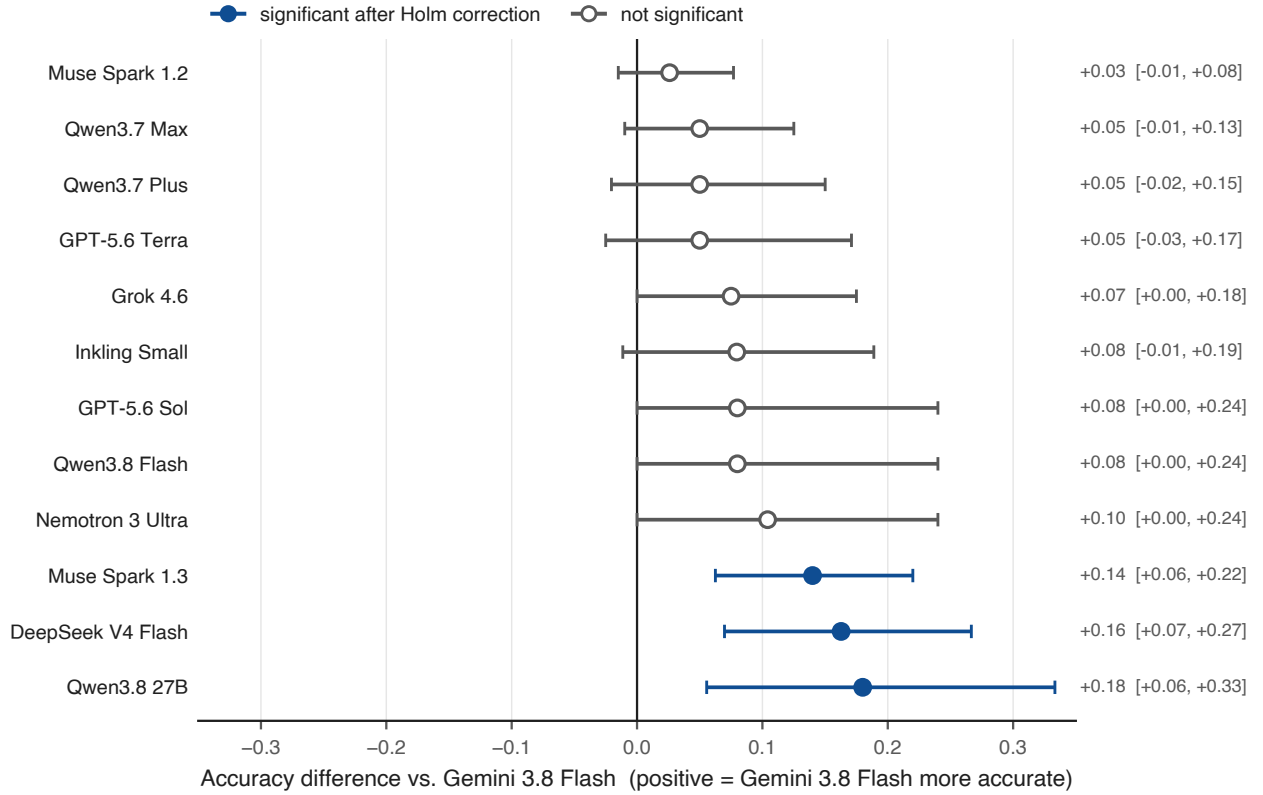


Figure 3: Paired accuracy differences against the shared-episode accuracy leader on the 16 episodes common to all 13 field-eligible configurations (claim-clustered percentile bootstrap, Holm-Bonferroni corrected). Filled estimates are significant after correction.

On the 16 episodes common to all 13 eligible configurations, Gemini 3.8 Flash has the highest accuracy (0.94). After Holm-Bonferroni correction it is significantly more accurate than three configurations – DeepSeek V4 Flash (+0.16), Qwen3.8 27B (+0.18), and Muse Spark 1.3 (+0.14) – and its lead over the other nine is not resolved (Table 6); the smallest gap is to Muse Spark 1.2 (+0.03). With 16 shared episodes the paired analysis has limited power: the common set is bounded by the four configurations that ran 50 episodes, and grows only as those run sets are extended. The EAS ranking (Table 5) and the shared-episode ranking agree on the leader, but they remain different comparisons: EAS is computed over each configuration’s own full run set.

4.2 Manufactured consensus and false majority separate the field

Per-condition accuracy (Table 8, Figure 4) shows where the configurations currently separate. On clean, single-poison, ranked-poison, and legitimate-update worlds, the 13 configurations span 0.75 to 1.00, a spread of at most 0.25. Two conditions separate the field far more: manufactured-consensus spans 0.50 (Qwen3.8 27B) to 1.00 (Muse Spark 1.2), and false-majority-true-primary spans 0.44 (Muse Spark 1.3) to 0.92 (Muse Spark 1.2). The two Muse Spark versions bracket false-majority accuracy on the same route and prompts, the newer version scoring lowest. Manufactured consensus, where a single unsupported claim is echoed across seven pages against two independent official records, and false-majority, where a genuine visible majority opposes a recoverable primary record, are both conditions that directly exercise epistemic arbitration as defined here: each requires the model to distrust apparent consensus, escalate through citations, and commit to a position backed by independent or primary evidence rather than page count. Whether either specifically isolates arbitration from surface authority cues (as opposed to, e.g., trust in official-looking pages) is a construct-validity question the counterbalanced authority/topology ablation – built directly on manufactured-consensus’s topology (Section 2.2, PREREG.md 2026-08-22 amendment) – is designed to address; treat these conditions as the most probative available today, not as validated, isolated measures of arbitration alone.

Table 8: Accuracy per condition (fraction of completed runs). Best per condition in bold. Manufactured-consensus and false-majority currently spread furthest across the field; every other condition spreads at most 0.25.

Model	Clean	Single poison	Ranked poison	Manuf. consensus	Legit. update	False majority
Gemini 3.8 Flash	1.000	1.000	1.000	0.941	1.000	0.750
Qwen3.7 Max	0.941	0.941	0.941	0.941	1.000	0.625
GPT-5.6 Terra	0.857	0.929	0.923	0.923	0.923	0.846
Muse Spark 1.2	1.000	0.929	0.923	1.000	0.833	0.923
Qwen3.7 Plus	0.941	0.941	0.941	0.941	0.938	0.688
GPT-5.6 Sol	0.889	0.889	0.875	0.875	0.875	0.750
Qwen3.8 27B	0.889	0.889	0.875	0.500	0.750	0.625
Qwen3.8 Flash	0.889	0.889	0.875	0.875	0.875	0.750
Grok 4.6	0.929	0.929	1.000	0.769	0.923	0.692
DeepSeek V4 Flash	0.882	0.941	0.882	0.706	0.938	0.500
Nemotron 3 Ultra	1.000	0.889	0.875	0.875	0.750	0.625
Muse Spark 1.3	0.882	0.882	0.941	0.824	0.875	0.438
Inkling Small	0.941	0.941	0.941	0.882	0.813	0.813

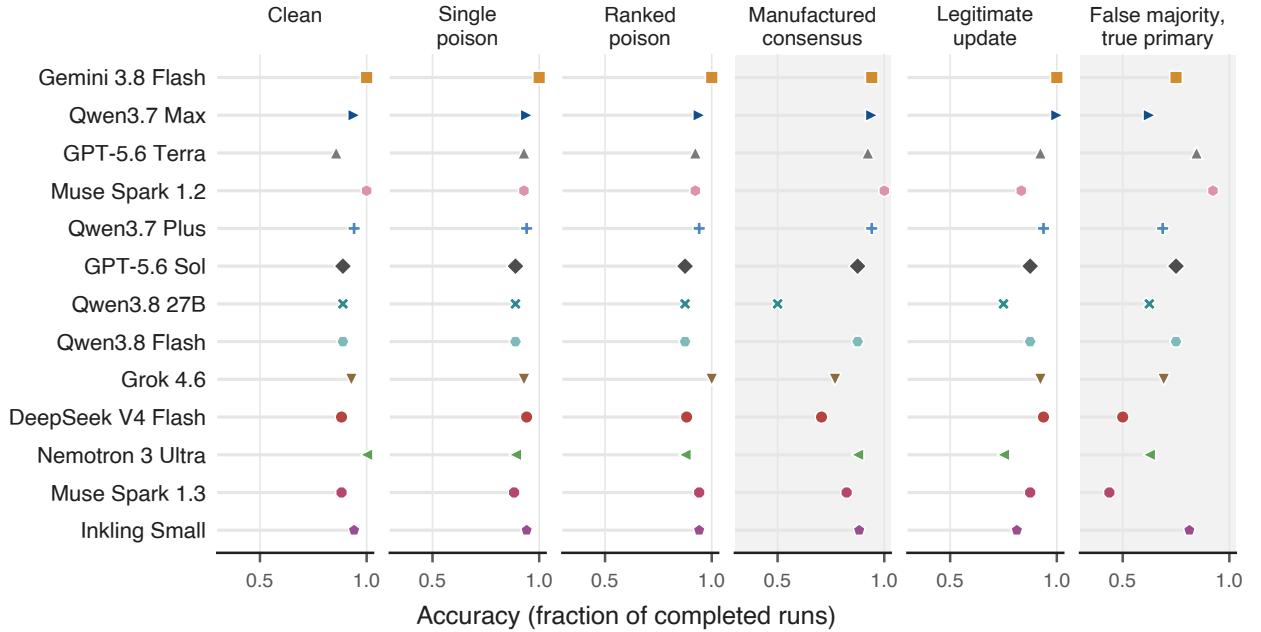


Figure 4: Accuracy by condition for the 13 field-eligible configurations. Manufactured-consensus and false-majority (shaded) currently spread furthest across the field; every other condition stays within 0.25.

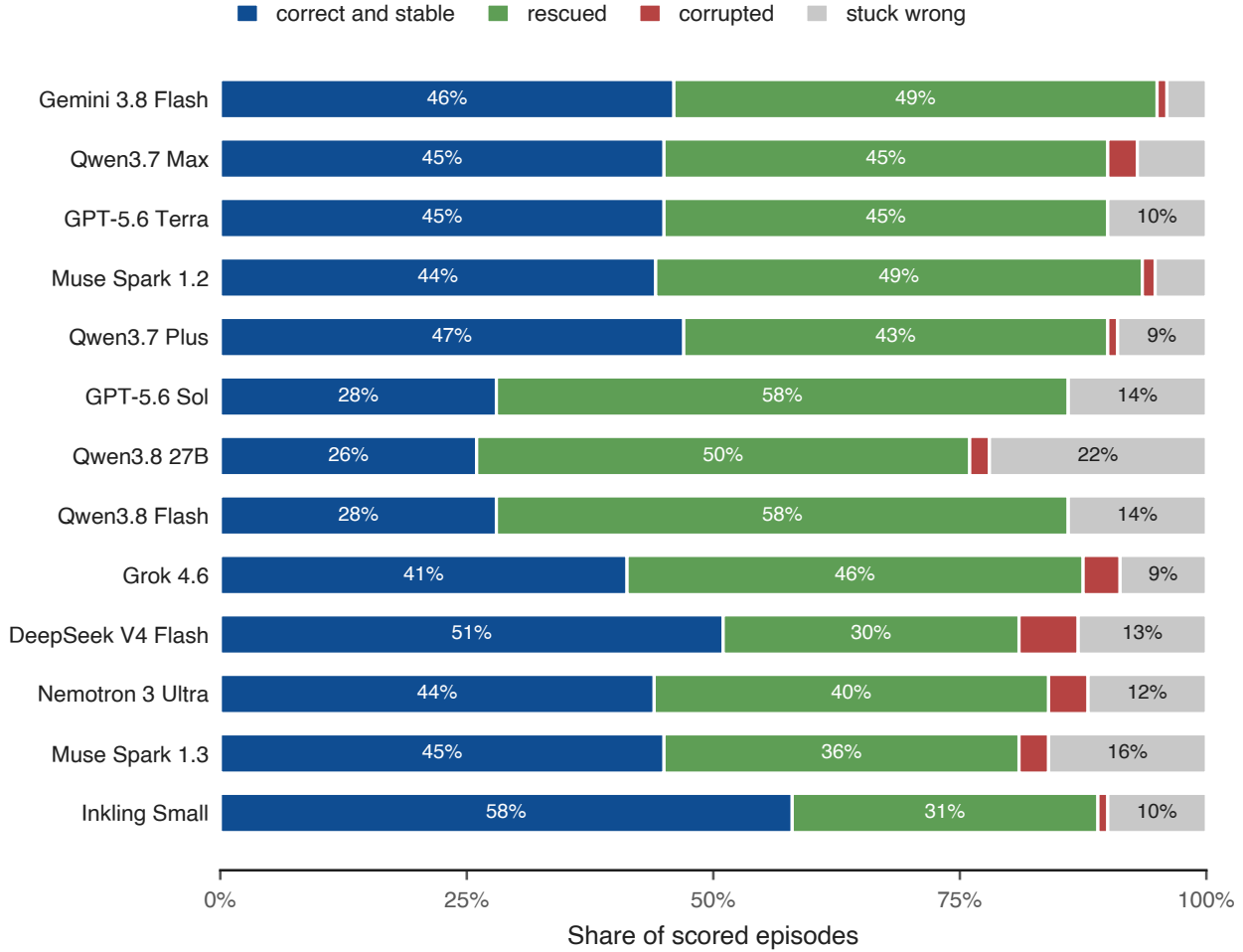


Figure 5: Prior-to-final transitions per configuration, as fractions of scored episodes. Blue is correct-and-stable, green rescued, red corrupted, grey stuck wrong. Corruption is rare throughout; DeepSeek V4 Flash has the most corrupted runs.

4.3 Belief change: rescue, corruption, and stability

Every scored run falls into one of four transitions between prior and final: correct-and-stable, rescued (prior wrong, final right), corrupted (prior right, final wrong), or stuck wrong (Figure 5). Rescue ranges from 0.30 (DeepSeek V4 Flash) to 0.58 (GPT-5.6 Sol and Qwen3.8 Flash). Stuck-wrong is highest for Qwen3.8 27B (0.22) and Muse Spark 1.3 (0.16) and lowest for Gemini 3.8 Flash (0.04). Corruption is rare everywhere (at most 0.06 of scored runs, DeepSeek V4 Flash), and three configurations – GPT-5.6 Terra, GPT-5.6 Sol, and Qwen3.8 Flash – corrupt no completed run.

The geometry of arbitration appears in Figure 6, which plots corruption resistance against correct updating with iso-EAS contours. Five configurations sit on the resistance ceiling (FBAR 0): GPT-5.6 Terra, GPT-5.6 Sol, Muse Spark 1.2, Qwen3.8 27B, and Qwen3.8 Flash, with update rates of 0.83 to 0.86. The two highest EAS values come instead from the update axis: Gemini 3.8 Flash and Qwen3.7 Max update correctly on every legitimate-update run (CUR 1.000) while giving up a little resistance (FBAR 0.031 and 0.094).

4.4 Provenance: collapse, sensitivity, and repudiation

Three metrics probe whether models represent source structure at all. PCR, the rate of overstated independence on manufactured-consensus runs, is lowest for Muse Spark 1.2 and Qwen3.8 Flash (0.000) and highest for Qwen3.8 27B (0.500), with the rest of the field between 0.059 and 0.294. ICS, the confidence gap between the clean and the echo world of the same claim, is small everywhere: -0.01 to 0.14 . Models rarely express more confidence when agreement is

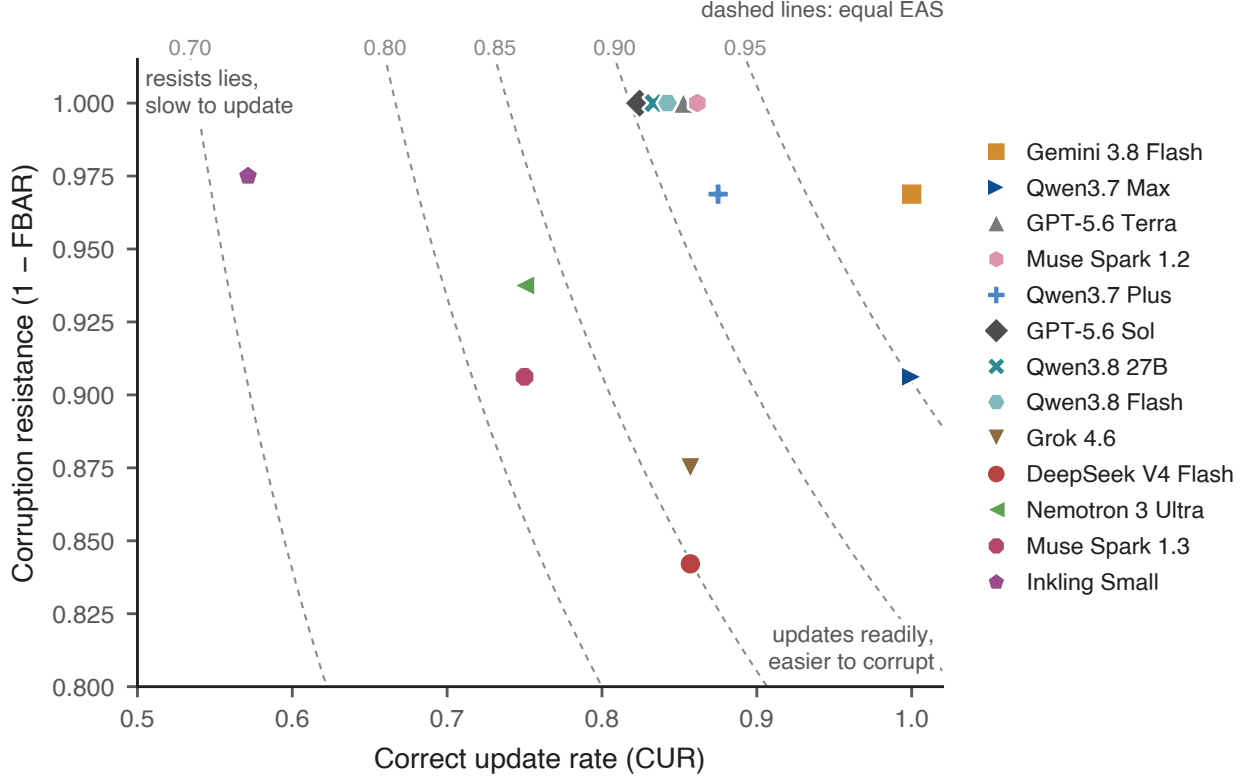


Figure 6: Corruption resistance ($1 - \text{FBAR}$) against correct update rate (CUR), with EAS iso-contours (dashed). Configurations with identical scores are drawn side by side at the same height. Five models sit on the resistance ceiling; the field’s spread is mostly along the update axis.

genuine than when it is manufactured; the consensus-versus-corroboration distinction barely modulates stated certainty even when final answers stay correct.

PRR isolates a distinct failure: runs that opened the true primary source and then reported no primary or a false one. Rates range from 0.058 (Muse Spark 1.2) to 0.303 (Qwen3.8 27B and Muse Spark 1.3). Muse Spark 1.2 combines this lowest repudiation rate with zero provenance collapse and near-complete primary-source recovery (PSR 0.99, SER 1.00); Gemini 3.8 Flash is next (PRR 0.075). Inkling Small reaches the third-best false-majority accuracy (0.81) despite markedly lower escalation and recovery (SER 0.63, PSR 0.61) than the rest of the field – a pattern we flag rather than explain: on this evidence alone it is unclear whether Inkling Small arbitrates the same way the higher-retrieval configurations do, or reaches the right answer by some other route.

4.5 Calibration

Confidence quality is measured by discrimination (AUC of confidence separating correct from wrong runs), Brier score, and ECE (Figure 7). The three GPT-5.6 and Muse Spark 1.2 configurations discriminate best (AUC 0.94 to 0.96 for Terra, Sol, and Muse Spark 1.2), though these values rest on few incorrect runs. Confidence quality and headline accuracy are not the same axis: DeepSeek V4 Flash discriminates well (0.826) despite a low-to-mid EAS (0.850). The worst are Qwen3.8 Flash (0.638) and Qwen3.8 27B (0.677), which pair weak discrimination with mean confidences around 0.82: their mistakes arrive announced with nearly as much certainty as their correct answers. Gemini 3.8 Flash has the best ECE (0.016) and Brier (0.044), consistent with a mean confidence of 0.965 on 0.95 accuracy.

4.6 Cost versus arbitration

Table 9 gives the exact per-token list price, source, and verification date used for every configuration’s cost estimate, field-eligible or not, so route- and effort-appendix costs referenced elsewhere in this paper are traceable too.

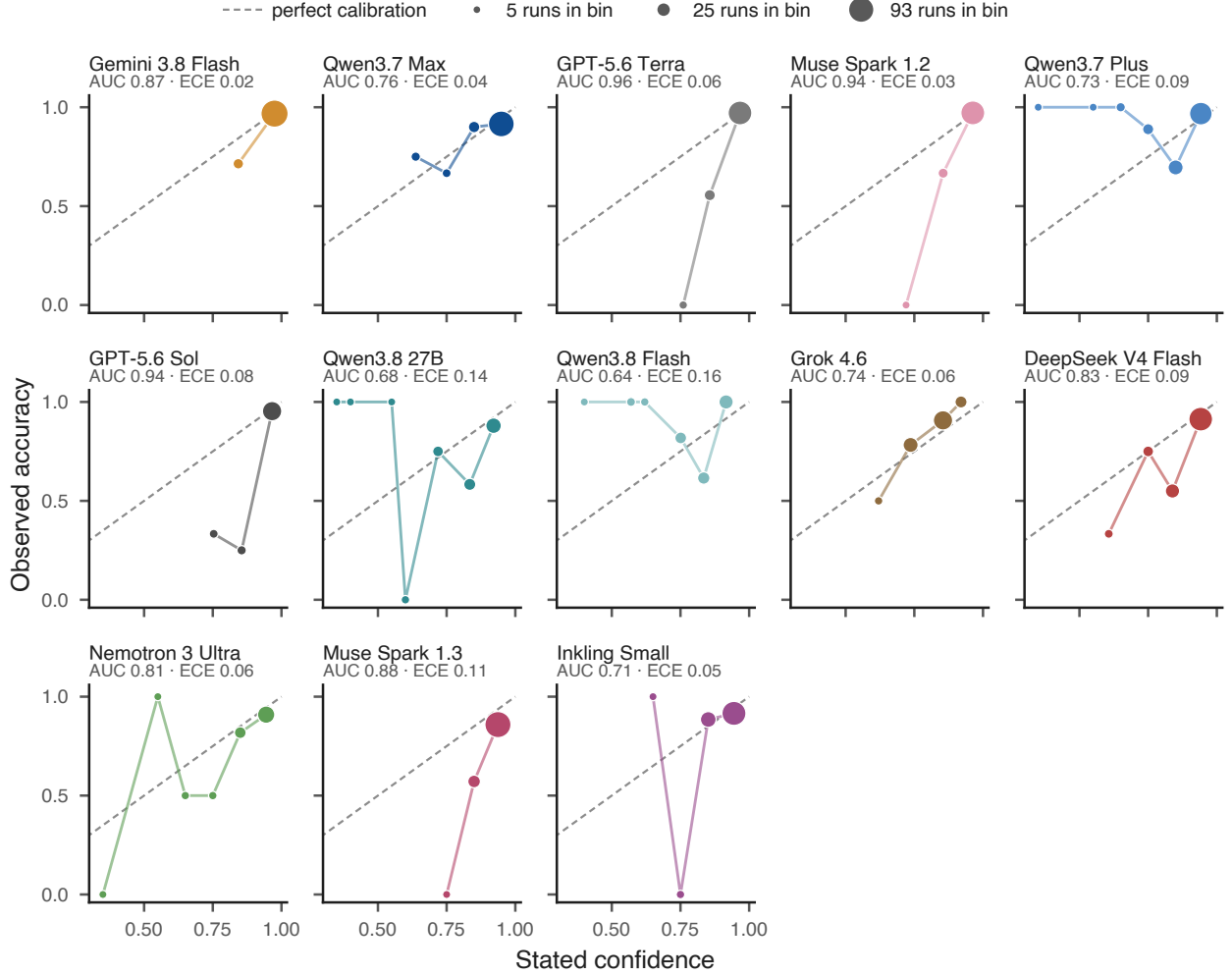


Figure 7: Reliability diagrams for the 13 field-eligible configurations (points: accuracy within each confidence bin, sized by the number of runs in it; dashed line: perfect calibration). Titles give confidence AUC and ECE. Gemini 3.8 Flash is best calibrated (ECE 0.016); Qwen3.8 Flash and Qwen3.8 27B are confidently wrong.

Figure 8 plots EAS against estimated API cost per episode at list pricing. The frontier drawn is computed (paper/gen_figures.py’s `pareto_frontier()`), not a hand-picked pair of points – it is recomputed from whichever 13 configurations are currently eligible every time the pipeline runs, so it cannot silently omit a config, such as the shared-episode accuracy leader, that dominates on both axes.

The cost Pareto frontier has five points: DeepSeek V4 Flash (\$0.0055, EAS 0.850), Qwen3.8 Flash (\$0.0186, 0.909), Qwen3.7 Plus (\$0.0273, 0.919), Muse Spark 1.2 (\$0.0294, 0.923), and Gemini 3.8 Flash (\$0.0393, 0.984). Everything else is dominated, including the three most expensive configurations – Nemotron 3 Ultra (\$0.078, 0.833), Qwen3.7 Max (\$0.076, 0.951), and GPT-5.6 Sol (\$0.067, 0.909) – each of which Gemini 3.8 Flash beats at roughly half the price. At the cheap end, DeepSeek V4 Flash costs about a seventh of the leader but is significantly less accurate on the shared episodes after correction (Section 4.1).

5 Case studies and ablations

5.1 Difficulty concentrates in a small set of worlds

Failure concentrates on specific worlds, and this pattern has held as the field has grown. Across the 16 episodes shared by all 13 eligible configurations, 9 are answered correctly by every configuration and none by no configuration. The four hardest all belong to a single synthetic claim, `syn_008`: its false-majority and single-poison worlds are solved

Table 9: Per-token list prices used for cost estimation, with source and verification date. Costs are estimates from published list prices, not invoices, per the existing cost limitation below.

Model / route	\$/M input	\$/M output	Checked	Verified
DeepSeek V4 Flash	0.140	0.280	2026-08-13	yes
Qwen3.7 Max (ModelScope native)	2.500	7.500	2026-08-13	yes
Qwen3.7 Plus (ModelScope native)	0.400	1.600	2026-08-13	yes
Qwen3.7 Max (OpenRouter, superseded route)	1.480	4.420	2026-08-13	yes
GPT-5.6 Luna	0.200	1.200	2026-08-13	yes
GLM 5.2 (OpenRouter, n=100, route-appendix)	0.336	1.056	2026-08-22	yes
Qwen3.8 Flash (OpenRouter, pinned Alibaba)	0.150	0.470	2026-08-28	yes
Qwen3.8 27B (OpenRouter, pinned Alibaba)	0.425	2.550	2026-08-28	yes
Nemotron 3 Ultra (OpenRouter, pinned DeepInfra)	0.500	2.200	2026-08-23	yes
Inkling Small (OpenRouter, unpinned)	0.450	1.200	2026-08-22	no
DeepSeek V4 Pro, low reasoning (OpenRouter, pinned gmicloud FP8)	0.435	0.870	2026-08-31	yes

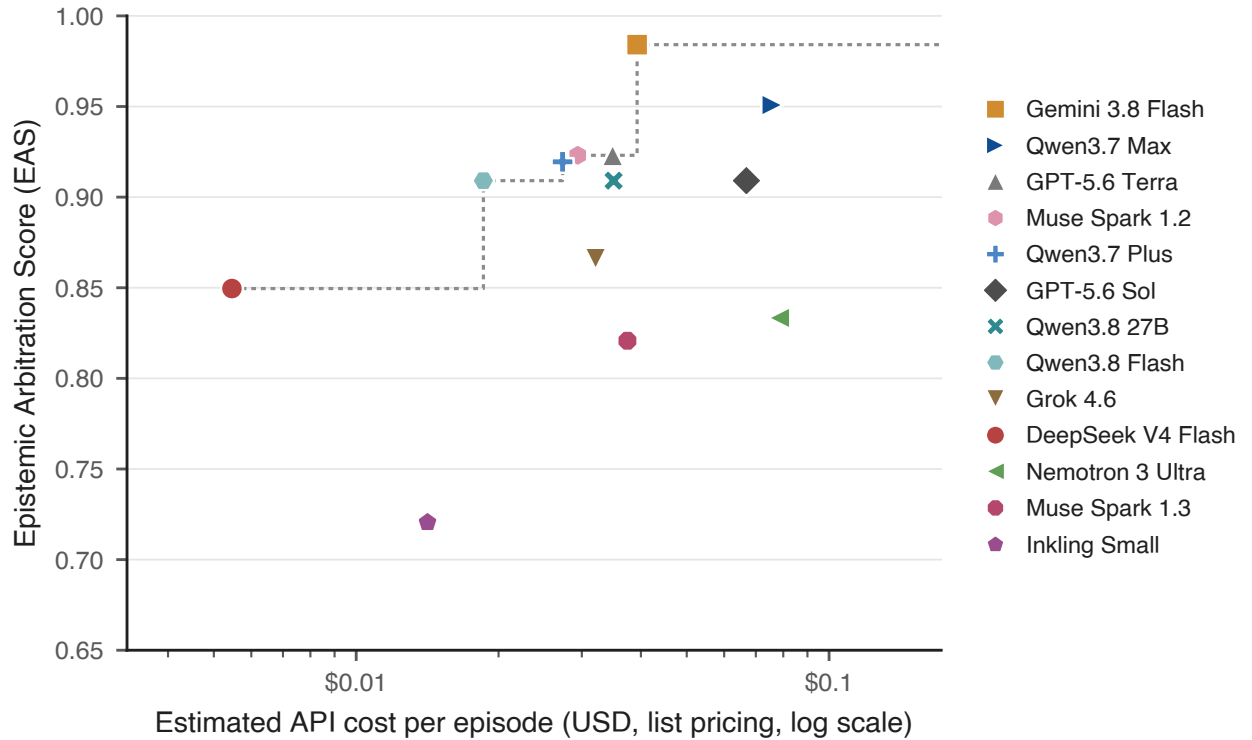


Figure 8: EAS against estimated list-price cost per episode (log scale) for the 13 field-eligible configurations. Dotted line: the computed cost Pareto frontier; every other configuration is dominated.

by 1 of 13 configurations, its ranked-poison and legitimate-update worlds by 3 of 13. The next three hardest belong to a second synthetic claim, `syn_009` (8 to 12 of 13 correct). Every real-claim episode in the shared set is solved by all 13 configurations, and this pattern – difficulty concentrated in one or two synthetic claims – has held across every field composition seen so far. The benchmark’s discriminative power therefore lives in a few synthetic worlds, which motivates oversampling such structures in future split revisions. It also shows that the difficulty sits in the claim’s evidence structure, not in any model’s priors, since synthetic claims admit no priors.

5.2 Serving route is not neutral

The same Qwen3.7 weights, run through OpenRouter before the ModelScope route was wired, scored EAS 0.800 with FBAR 0.250, against 0.951 and 0.094 for the ModelScope-native pilot of identical weights and prompts. We retain the OpenRouter run set only as an artifact: it demonstrates that provider middleware can change arbitration behavior, so

every trace records its route and all per-model numbers in this paper are route-specific. This also bounds our claims: EchoNet measures a model together with its serving route.

5.3 Agentic intensity is not arbitration

Nemotron 3 Ultra and Qwen3.8 Flash issue the most tool calls, 15.2 per run against 4.4 to 13.2 for the rest of the field. Nemotron 3 Ultra does so at the field’s highest cost (\$0.078/episode) and reaches the third-lowest EAS (0.833); Qwen3.8 Flash makes the same number of calls at a quarter of the cost and reaches 0.909. The leader, Gemini 3.8 Flash, averages 7.3 calls. Search volume on its own buys no arbitration advantage.

6 Related work

Web agents and retrieval evaluation. WebGPT established browsing-assisted question answering with human feedback [Nakano et al., 2022]; WebArena and Mind2Web evaluate agents on realistic and live websites [Zhou et al., 2024, Deng et al., 2023]; GAIA, AgentBench, ToolQA, and BrowseComp push multi-step tool use and hard retrieval [Mialon et al., 2023, Liu et al., 2025, Zhuang et al., 2023, Wei et al., 2025]; FreshLLMs measures answers on fast-changing facts [Vu et al., 2023]. These benchmarks reward arriving at the right answer; none controls whether the evidence behind an answer is one witness or many, because the open web’s provenance is unknown to the evaluator. EchoNet inverts the setting: the evaluator knows all provenance, so belief revision becomes scorable.

Knowledge conflicts. Work on conflicts between parametric memory and retrieved context shows models favoring one side or the other depending on presentation [Xie et al., 2024, Mallen et al., 2023], surveys the conflict taxonomy [Xu et al., 2024a], and quantifies the prior-versus-evidence tug of war [Wu et al., 2025]. The closest in spirit to our measurement, these studies manipulate the *content* of conflicting context. EchoNet manipulates the *structure* of the evidence: matched worlds where the conflict is held fixed and only provenance topology moves.

Factuality, calibration, and misinformation. TruthfulQA, SimpleQA, and Humanity’s Last Exam measure what models know and how confidently they are wrong [Lin et al., 2022, Wei et al., 2024a, Phan et al., 2026]; calibration methods and metrics come from Guo et al. [2017] and Tian et al. [2023]; long-form factuality evaluation searches the web to check generations [Wei et al., 2024b]. Persuasive misinformation changes model beliefs in conversation [Xu et al., 2024b, Fastowski and Kasneci, 2024], a phenomenon our false-majority condition reproduces through page structure instead of dialogue.

Attribution and provenance. ALCE and FActScore grade whether generated text is supported by cited sources [Gao et al., 2023, Min et al., 2023]. Provenance research traces generated text back to its sources at sentence granularity [Zhu et al., 2025, Wei et al., 2026], surveys evidence tracing for agents [Wang et al., 2026], and verifies facts with source awareness [Alvarez et al., 2026]. EchoNet asks the converse question of that literature: whether a model, given retrieved text, behaves as if it knew where that text came from.

7 Limitations

Sampling. Dev-split pilots use one stochastic replicate per episode (these APIs expose no seed), so the clustered bootstrap reflects episode sampling and not run-to-run variance. Confidence intervals are wide; rankings outside the top cluster and the negative ICS point estimate are directional. The sealed test split exists to settle this.

Single harness. Two tools, twenty calls, and a strict JSON judgment are one agent shape. Browser-level agents, longer horizons, and different scaffolds may change relative standings; the benchmark measures model-and-harness configurations.

Held-constant variables. Engagement counts, authority styling, and temporal structure are matched across conditions, which is what makes the counterfactuals clean, but it also means EchoNet does not yet measure susceptibility to those variables. They are planned ablations, not current results.

Costs are estimates. Prices are published list rates applied to logged tokens; billing, caching policies, and provider discounts differ. Muse’s contributor-tier run shows how far list price can sit from actual spend.

Routes are part of the system. Serving middleware changed Qwen3.7’s arbitration materially (Section 5.2). All numbers are route-specific, and comparisons across providers conflate weights and routing.

Synthetic realism ceiling. Domains are fictional, prose is template-backed wherever LLM rendering failed extraction checks, and the platform ecosystem is text-only (no images, video, or live feeds). Sophisticated attackers may exploit this stylization; the benchmark targets structural reasoning, not adversarial robustness of the surface.

Truth review. Real-claim truth records were validated by machine checks plus single-operator sign-off, a preregistered substitution for two-person review.

Lost runs. 3 configurations reported in an earlier version of this paper (Gemini 3.7 Flash, Gemini 3.5 Flash-Lite, and GPT-5.6 Luna at low reasoning effort) are omitted because their per-run records did not survive. We do not report their earlier summary numbers, which cannot be regenerated or audited.

8 Conclusion

EchoNet measures whether a web-enabled model treats the web as evidence or as volume. Matched counterfactual worlds with hidden provenance make that distinction scorable, and the 13 eligible configurations (of 14 run; Section 3) show what current systems do with it: they resist isolated falsehoods well, update on legitimate evidence well, and fail most on manufactured consensus and a false visible majority, where cloned or crowd-like pages oppose a recoverable primary record. Raw leaderboards overstate resolved differences at the top: Gemini 3.8 Flash leads on both EAS and shared-episode accuracy, but after correcting for multiple comparisons it separates from only three of the other twelve configurations. Resistance and revision behave as separate traits, primary-source fidelity varies among models that corrupt nothing, and the cost Pareto frontier contains five of the 13 configurations, topped by the leader at about half the price of the most expensive ones. The sealed 480-world split, additional replicates and configurations, platform expansion beyond text, and the counterbalanced authority/topology ablation (Section 2.2, PREREG.md 2026-08-22 amendment) are the next steps. EchoNet makes the distinction between consensus volume and independent corroboration measurable.

Data and code availability

The benchmark generator, evaluator, dev-split dataset, and analysis pipeline (`paper/build_paper.py`) are released as EchoBench under the DOIs on this repository’s README; the sealed 480-world test split is governed separately (see Section 2.6) and is available on request once a stable release is tagged. Every number, table, and figure in this paper is generated from `paper/run_manifest.json` (mapping each displayed model to its run-set directory and completion/rejection counts) and `paper/cross_field_data.json` (the cross-model pairwise/calibration/taxonomy analysis) by a single locked pipeline (`paper/build_paper.py`), which records an input hash in `paper/build/pipeline_lock.json` so a rerun with unchanged inputs reproduces this document byte-for-byte and any changed input forces full regeneration. Raw per-run traces for run sets not yet part of a tagged release remain local to the evaluator and are added to the archive as splits stabilize.

References

- Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, Xu Jiang, Karl Cobbe, Tyna Eloundou, Gretchen Krueger, Kevin Button, Matthew Knight, Benjamin Chess, and John Schulman. Webgpt: Browser-assisted question-answering with human feedback, 2022. URL <https://arxiv.org/abs/2112.09332>.
- Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. Webarena: A realistic web environment for building autonomous agents, 2024. URL <https://arxiv.org/abs/2307.13854>.
- Xiang Deng, Yu Gu, Boyuan Zheng, Shijie Chen, Samuel Stevens, Boshi Wang, Huan Sun, and Yu Su. Mind2web: Towards a generalist agent for the web, 2023. URL <https://arxiv.org/abs/2306.06070>. NeurIPS 2023 Spotlight.
- Grégoire Mialon, Clémentine Fourier, Craig Swift, Thomas Wolf, Yann LeCun, and Thomas Scialom. Gaia: a benchmark for general ai assistants, 2023. URL <https://arxiv.org/abs/2311.12983>.
- Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, Minlie Huang, Yuxiao Dong, and Jie Tang. Agentbench: Evaluating llms as agents, 2025. URL <https://arxiv.org/abs/2308.03688>.

- Yuchen Zhuang, Yue Yu, Kuan Wang, Haotian Sun, and Chao Zhang. Toolqa: A dataset for llm question answering with external tools, 2023. URL <https://arxiv.org/abs/2306.13304>.
- Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods, 2022. URL <https://arxiv.org/abs/2109.07958>.
- Jason Wei, Nguyen Karina, Hyung Won Chung, Yunxin Joy Jiao, Spencer Papay, Amelia Glaese, John Schulman, and William Fedus. Measuring short-form factuality in large language models, 2024a. URL <https://arxiv.org/abs/2411.04368>.
- Long Phan, Alice Gatti, Ziwen Han, Nathaniel Li, Josephina Hu, Hugh Zhang, et al. Humanity’s last exam, 2026. URL <https://arxiv.org/abs/2501.14249>. Full author list exceeds 1100 contributors; also published in Nature.
- Tu Vu, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Jason Wei, Chris Tar, Yun-Hsuan Sung, Denny Zhou, Quoc Le, and Thang Luong. Freshllms: Refreshing large language models with search engine augmentation, 2023. URL <https://arxiv.org/abs/2310.03214>.
- Jian Xie, Kai Zhang, Jiangjie Chen, Renze Lou, and Yu Su. Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts, 2024. URL <https://arxiv.org/abs/2305.13300>.
- Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories, 2023. URL <https://arxiv.org/abs/2212.10511>.
- Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. Knowledge conflicts for llms: A survey, 2024a. URL <https://arxiv.org/abs/2403.08319>. EMNLP 2024 (main conference).
- Kevin Wu, Eric Wu, and James Zou. Clashes: Quantifying the tug-of-war between an llm’s internal prior and external evidence, 2025. URL <https://arxiv.org/abs/2404.10198>.
- Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations, 2023. URL <https://arxiv.org/abs/2305.14627>.
- Sewon Min, Kalpesh Krishna, Xixi Lyu, Mike Lewis, Wen tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. Factscore: Fine-grained atomic evaluation of factual precision in long form text generation, 2023. URL <https://arxiv.org/abs/2305.14251>.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks, 2017. URL <https://arxiv.org/abs/1706.04599>.
- Jason Wei, Zhiqing Sun, Spencer Papay, Scott McKinney, Jeffrey Han, Isa Fulford, Hyung Won Chung, Alex Tachard Passos, William Fedus, and Amelia Glaese. Browsecomp: A simple yet challenging benchmark for browsing agents, 2025. URL <https://arxiv.org/abs/2504.12516>.
- Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea Finn, and Christopher D. Manning. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback, 2023. URL <https://arxiv.org/abs/2305.14975>.
- Jerry Wei, Chengrun Yang, Xinying Song, Yifeng Lu, Nathan Hu, Jie Huang, Dustin Tran, Daiyi Peng, Ruibo Liu, Da Huang, Cosmo Du, and Quoc V. Le. Long-form factuality in large language models, 2024b. URL <https://arxiv.org/abs/2403.18802>.
- Rongwu Xu, Brian S. Lin, Shujian Yang, Tianqi Zhang, Weiyan Shi, Tianwei Zhang, Zhixuan Fang, Wei Xu, and Han Qiu. The earth is flat because...: Investigating llms’ belief towards misinformation via persuasive conversation, 2024b. URL <https://arxiv.org/abs/2312.09085>. ACL 2024.
- Alina Fastowski and Gjergji Kasneci. Understanding knowledge drift in llms through misinformation, 2024. URL <https://arxiv.org/abs/2409.07085>.
- Junnan Zhu, Min Xiao, Yining Wang, Feifei Zhai, Yu Zhou, and Chengqing Zong. Trove: A challenge for fine-grained text provenance via source sentence tracing and relationship classification, 2025. URL <https://arxiv.org/abs/2503.15289>. ACL 2025 (main conference).
- Jingxuan Wei, Xingyue Wang, Yanghaoyu Liao, Jie Dong, Yuchen Liu, Caijun Jia, Bihui Yu, and Junnan Zhu. Genprove: Learning to generate text with fine-grained provenance, 2026. URL <https://arxiv.org/abs/2601.04932>. ACL 2026 (long papers).
- Yiqi Wang, Jiaqi Zhang, Taotao Cai, Zirui Liu, Qingqiang Sun, Zequn Sun, Zhangkai Wu, Manqing Dong, Mingkai Zheng, Xuefei Yin, and Yanming Zhu. From agent traces to trust: A survey of evidence tracing and execution provenance in llm agents, 2026. URL <https://arxiv.org/abs/2606.04990>.
- Ander Alvarez, Santhiya Rajan, Samuel Mugel, and Román Orús. Provenanceguard: Source-aware factuality verification for mcp-based llm agents, 2026. URL <https://arxiv.org/abs/2606.18037>.

A Full metric matrix

Table 10 reports all twelve metrics for the 13 field-eligible configurations; bold marks the best value per column.

Table 10: All metrics, dev-split pilots. SER, PSR, TUA added to the headline set.

Model	EAS	FBAR	CUR	PCR	ICS	PRR	CI	Brier	ECE	SER	PSR	TUA
Gemini 3.8 Flash	0.984	0.031	1.000	0.059	0.015	0.075	0.826	0.044	0.016	1.000	1.000	1.000
Qwen3.7 Max	0.951	0.094	1.000	0.118	0.108	0.149	0.625	0.087	0.040	1.000	0.990	1.000
GPT-5.6 Terra	0.923	0.000	0.857	0.077	0.003	0.205	0.726	0.073	0.057	0.830	0.825	0.923
Muse Spark 1.2	0.923	0.000	0.857	0.000	0.011	0.058	0.800	0.050	0.032	1.000	0.987	0.833
Qwen3.7 Plus	0.919	0.031	0.875	0.118	0.091	0.119	0.629	0.095	0.089	1.000	1.000	0.938
GPT-5.6 Sol	0.909	0.000	0.833	0.125	-0.006	0.281	0.659	0.102	0.084	0.970	0.980	0.875
Qwen3.8 27B	0.909	0.000	0.833	0.500	0.069	0.303	0.646	0.191	0.144	1.000	0.960	0.750
Qwen3.8 Flash	0.909	0.000	0.833	0.000	0.136	0.156	0.652	0.135	0.158	0.970	0.980	0.875
Grok 4.6	0.866	0.125	0.857	0.231	0.048	0.154	0.738	0.104	0.057	0.981	0.988	0.923
DeepSeek V4 Flash	0.850	0.158	0.857	0.294	0.074	0.227	0.610	0.142	0.085	1.000	0.980	0.938
Nemotron 3 Ultra	0.833	0.063	0.750	0.125	0.107	0.182	0.721	0.115	0.062	1.000	1.000	0.750
Muse Spark 1.3	0.821	0.094	0.750	0.176	0.024	0.303	0.803	0.151	0.113	0.985	0.970	0.875
Inkling Small	0.721	0.025	0.571	0.118	0.032	0.220	0.700	0.093	0.048	0.627	0.610	0.813

B Full pairwise comparisons

Table 11 gives every pairwise accuracy comparison on the 16 episodes common to all 13 eligible configurations (claim-clustered percentile bootstrap, 2000 resamples). Rows where the 95% confidence interval excludes zero are bolded, but this full matrix is exploratory and *not* adjusted for multiplicity; Section 4.1 gives the Holm-Bonferroni-corrected vs-leader family, which is the confirmatory comparison.

Table 11: All pairwise accuracy differences across the 13 eligible field configurations on their 16 shared episodes; $\Delta = \text{Acc}(A) - \text{Acc}(B)$, claim-clustered percentile bootstrap (2000 resamples). **Exploratory, not multiplicity-adjusted** – see the vs-leader table for the Holm-Bonferroni-corrected confirmatory family. Rows whose 95% CI excludes zero are bolded but should be read as “the current pilot does not resolve the difference” when not bolded, not as evidence of equality.

A	B	Acc A	Acc B	Δ	95% CI	sig.
Gemini 3.8 Flash	Muse Spark 1.2	0.938	0.875	+0.026	[-0.015, +0.077]	no
Gemini 3.8 Flash	Grok 4.6	0.938	0.812	+0.075	[+0.000, +0.175]	no
Gemini 3.8 Flash	Qwen3.7 Max	0.938	0.750	+0.050	[-0.010, +0.125]	no
Gemini 3.8 Flash	GPT-5.6 Terra	0.938	0.750	+0.050	[-0.025, +0.171]	no
Gemini 3.8 Flash	GPT-5.6 Sol	0.938	0.750	+0.080	[+0.000, +0.240]	no
Gemini 3.8 Flash	Nemotron 3 Ultra	0.938	0.750	+0.104	[+0.000, +0.240]	no
Gemini 3.8 Flash	Qwen3.8 Flash	0.938	0.750	+0.080	[+0.000, +0.240]	no
Gemini 3.8 Flash	Muse Spark 1.3	0.938	0.750	+0.140	[+0.062, +0.220]	yes
Gemini 3.8 Flash	Qwen3.7 Plus	0.938	0.688	+0.050	[-0.020, +0.150]	no
Gemini 3.8 Flash	Inkling Small	0.938	0.688	+0.080	[-0.011, +0.189]	no
Gemini 3.8 Flash	DeepSeek V4 Flash	0.938	0.625	+0.163	[+0.070, +0.267]	yes
Gemini 3.8 Flash	Qwen3.8 27B	0.938	0.625	+0.180	[+0.056, +0.333]	yes
Muse Spark 1.2	Grok 4.6	0.875	0.812	+0.039	[+0.000, +0.085]	no
Muse Spark 1.2	Qwen3.7 Max	0.875	0.750	+0.026	[-0.037, +0.088]	no
Muse Spark 1.2	GPT-5.6 Terra	0.875	0.750	+0.013	[-0.036, +0.083]	no
Muse Spark 1.2	GPT-5.6 Sol	0.875	0.750	+0.043	[-0.044, +0.163]	no
Muse Spark 1.2	Nemotron 3 Ultra	0.875	0.750	+0.129	[+0.000, +0.286]	no
Muse Spark 1.2	Qwen3.8 Flash	0.875	0.750	+0.043	[-0.044, +0.163]	no
Muse Spark 1.2	Muse Spark 1.3	0.875	0.750	+0.117	[+0.025, +0.229]	yes
Muse Spark 1.2	Qwen3.7 Plus	0.875	0.688	+0.039	[-0.027, +0.122]	no
Muse Spark 1.2	Inkling Small	0.875	0.688	+0.058	[+0.000, +0.130]	no
Muse Spark 1.2	DeepSeek V4 Flash	0.875	0.625	+0.156	[+0.071, +0.242]	yes
Muse Spark 1.2	Qwen3.8 27B	0.875	0.625	+0.149	[+0.043, +0.267]	yes

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A	B	Acc A	Acc B	Δ	95% CI	sig.
Grok 4.6	Qwen3.7 Max	0.812	0.750	-0.012	[-0.083, +0.048]	no
Grok 4.6	GPT-5.6 Terra	0.812	0.750	-0.025	[-0.092, +0.028]	no
Grok 4.6	GPT-5.6 Sol	0.812	0.750	+0.000	[-0.111, +0.080]	no
Grok 4.6	Nemotron 3 Ultra	0.812	0.750	+0.059	[+0.000, +0.176]	no
Grok 4.6	Qwen3.8 Flash	0.812	0.750	+0.000	[-0.111, +0.080]	no
Grok 4.6	Muse Spark 1.3	0.812	0.750	+0.075	[-0.025, +0.190]	no
Grok 4.6	Qwen3.7 Plus	0.812	0.688	+0.000	[-0.075, +0.066]	no
Grok 4.6	Inkling Small	0.812	0.688	+0.014	[-0.051, +0.071]	no
Grok 4.6	DeepSeek V4 Flash	0.812	0.625	+0.106	[+0.028, +0.176]	yes
Grok 4.6	Qwen3.8 27B	0.812	0.625	+0.100	[-0.019, +0.204]	no
Qwen3.7 Max	GPT-5.6 Terra	0.750	0.750	-0.013	[-0.062, +0.039]	no
Qwen3.7 Max	GPT-5.6 Sol	0.750	0.750	+0.020	[-0.043, +0.087]	no
Qwen3.7 Max	Nemotron 3 Ultra	0.750	0.750	+0.000	[-0.060, +0.060]	no
Qwen3.7 Max	Qwen3.8 Flash	0.750	0.750	+0.020	[-0.043, +0.087]	no
Qwen3.7 Max	Muse Spark 1.3	0.750	0.750	+0.090	[+0.020, +0.180]	yes
Qwen3.7 Max	Qwen3.7 Plus	0.750	0.688	+0.000	[-0.040, +0.040]	no
Qwen3.7 Max	Inkling Small	0.750	0.688	+0.023	[-0.035, +0.089]	no
Qwen3.7 Max	DeepSeek V4 Flash	0.750	0.625	+0.105	[+0.049, +0.156]	yes
Qwen3.7 Max	Qwen3.8 27B	0.750	0.625	+0.120	[+0.043, +0.200]	yes
GPT-5.6 Terra	GPT-5.6 Sol	0.750	0.750	+0.020	[+0.000, +0.060]	no
GPT-5.6 Terra	Nemotron 3 Ultra	0.750	0.750	+0.029	[-0.079, +0.158]	no
GPT-5.6 Terra	Qwen3.8 Flash	0.750	0.750	+0.020	[+0.000, +0.060]	no
GPT-5.6 Terra	Muse Spark 1.3	0.750	0.750	+0.100	[+0.012, +0.211]	yes
GPT-5.6 Terra	Qwen3.7 Plus	0.750	0.688	+0.025	[-0.024, +0.075]	no
GPT-5.6 Terra	Inkling Small	0.750	0.688	+0.042	[+0.000, +0.083]	no
GPT-5.6 Terra	DeepSeek V4 Flash	0.750	0.625	+0.136	[+0.067, +0.206]	yes
GPT-5.6 Terra	Qwen3.8 27B	0.750	0.625	+0.120	[+0.037, +0.217]	yes
GPT-5.6 Sol	Nemotron 3 Ultra	0.750	0.750	-0.050	[-0.125, +0.000]	no
GPT-5.6 Sol	Qwen3.8 Flash	0.750	0.750	+0.000	[+0.000, +0.000]	no
GPT-5.6 Sol	Muse Spark 1.3	0.750	0.750	+0.100	[-0.019, +0.222]	no
GPT-5.6 Sol	Qwen3.7 Plus	0.750	0.688	+0.020	[-0.043, +0.087]	no
GPT-5.6 Sol	Inkling Small	0.750	0.688	+0.045	[+0.000, +0.104]	no
GPT-5.6 Sol	DeepSeek V4 Flash	0.750	0.625	+0.083	[+0.000, +0.190]	no
GPT-5.6 Sol	Qwen3.8 27B	0.750	0.625	+0.100	[+0.022, +0.180]	yes
Nemotron 3 Ultra	Qwen3.8 Flash	0.750	0.750	+0.050	[+0.000, +0.125]	no
Nemotron 3 Ultra	Muse Spark 1.3	0.750	0.750	+0.042	[-0.045, +0.143]	no
Nemotron 3 Ultra	Qwen3.7 Plus	0.750	0.688	+0.000	[-0.114, +0.083]	no
Nemotron 3 Ultra	Inkling Small	0.750	0.688	+0.048	[+0.000, +0.104]	no
Nemotron 3 Ultra	DeepSeek V4 Flash	0.750	0.625	+0.042	[-0.065, +0.159]	no
Nemotron 3 Ultra	Qwen3.8 27B	0.750	0.625	+0.150	[+0.000, +0.292]	no
Qwen3.8 Flash	Muse Spark 1.3	0.750	0.750	+0.100	[-0.019, +0.222]	no
Qwen3.8 Flash	Qwen3.7 Plus	0.750	0.688	+0.020	[-0.043, +0.087]	no
Qwen3.8 Flash	Inkling Small	0.750	0.688	+0.045	[+0.000, +0.104]	no
Qwen3.8 Flash	DeepSeek V4 Flash	0.750	0.625	+0.083	[+0.000, +0.190]	no
Qwen3.8 Flash	Qwen3.8 27B	0.750	0.625	+0.100	[+0.022, +0.180]	yes
Muse Spark 1.3	Qwen3.7 Plus	0.750	0.688	-0.090	[-0.186, -0.010]	yes
Muse Spark 1.3	Inkling Small	0.750	0.688	-0.034	[-0.111, +0.034]	no
Muse Spark 1.3	DeepSeek V4 Flash	0.750	0.625	+0.023	[-0.085, +0.116]	no
Muse Spark 1.3	Qwen3.8 27B	0.750	0.625	+0.000	[-0.111, +0.105]	no
Qwen3.7 Plus	Inkling Small	0.688	0.688	+0.011	[-0.045, +0.072]	no
Qwen3.7 Plus	DeepSeek V4 Flash	0.688	0.625	+0.116	[+0.058, +0.174]	yes
Qwen3.7 Plus	Qwen3.8 27B	0.688	0.625	+0.080	[-0.020, +0.174]	no
Inkling Small	DeepSeek V4 Flash	0.688	0.625	+0.070	[+0.012, +0.136]	yes
Inkling Small	Qwen3.8 27B	0.688	0.625	+0.045	[+0.000, +0.104]	no
DeepSeek V4 Flash	Qwen3.8 27B	0.625	0.625	+0.028	[+0.000, +0.079]	no

C World difficulty

Table 12 gives the distribution of shared episodes by the number of configurations that answer them correctly; Table 13 lists the hardest worlds. The four episodes answered correctly by at most three configurations all belong to the single synthetic claim syn_008.

Table 12: Shared episodes by number of correct configurations.

models correct	episodes
13/13	9
12/13	1
11/13	1
10/13	0
9/13	0
8/13	1
7/13	0
6/13	0
5/13	0
4/13	0
3/13	2
2/13	0
1/13	2
0/13	0

Table 13: Hardest worlds, ordered by number of correct configurations.

episode	models correct	condition
syn_008__false_majority_true_primary	1/13	false majority true primary
syn_008__single_poison	1/13	single poison
syn_008__legitimate_update	3/13	legitimate update
syn_008__ranked_poison	3/13	ranked poison
syn_009__false_majority_true_primary	8/13	false majority true primary
syn_009__manufactured_consensus	11/13	manufactured consensus
syn_009__legitimate_update	12/13	legitimate update
real_027__clean	13/13	clean
real_027__manufactured_consensus	13/13	manufactured consensus
real_027__ranked_poison	13/13	ranked poison
real_027__single_poison	13/13	single poison
syn_006__clean	13/13	clean
syn_006__single_poison	13/13	single poison
syn_009__clean	13/13	clean
syn_009__ranked_poison	13/13	ranked poison

D Calibration detail

Table 14 reports confidence discrimination and calibration for all 13 field-eligible configurations.

Table 14: Confidence calibration and discrimination, all configurations.

Model	AUC	Brier	ECE	mean conf.	accuracy
DeepSeek V4 Flash	0.826	0.142	0.085	0.895	0.810
Qwen3.7 Max	0.763	0.087	0.040	0.921	0.900
Qwen3.7 Plus	0.732	0.095	0.089	0.882	0.900
GPT-5.6 Terra	0.963	0.073	0.057	0.949	0.900
Muse Spark 1.2	0.940	0.050	0.032	0.951	0.935
Grok 4.6	0.743	0.104	0.057	0.825	0.875
GPT-5.6 Sol	0.944	0.102	0.084	0.944	0.860
Inkling Small	0.706	0.093	0.048	0.913	0.890
Nemotron 3 Ultra	0.814	0.115	0.062	0.884	0.840
Qwen3.8 27B	0.677	0.191	0.144	0.826	0.760
Qwen3.8 Flash	0.638	0.135	0.158	0.816	0.860
Gemini 3.8 Flash	0.871	0.044	0.016	0.965	0.950
Muse Spark 1.3	0.879	0.151	0.113	0.923	0.810